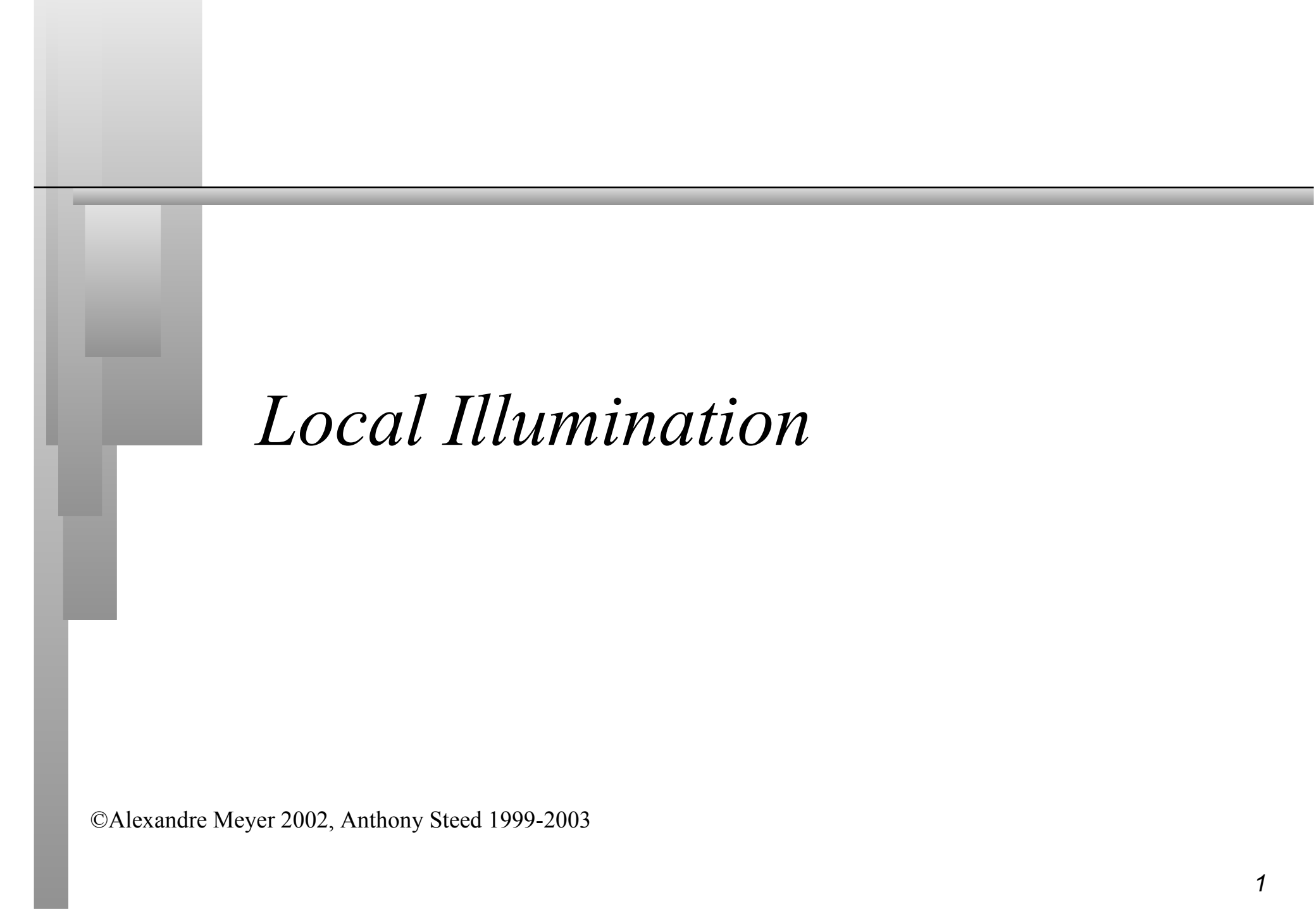
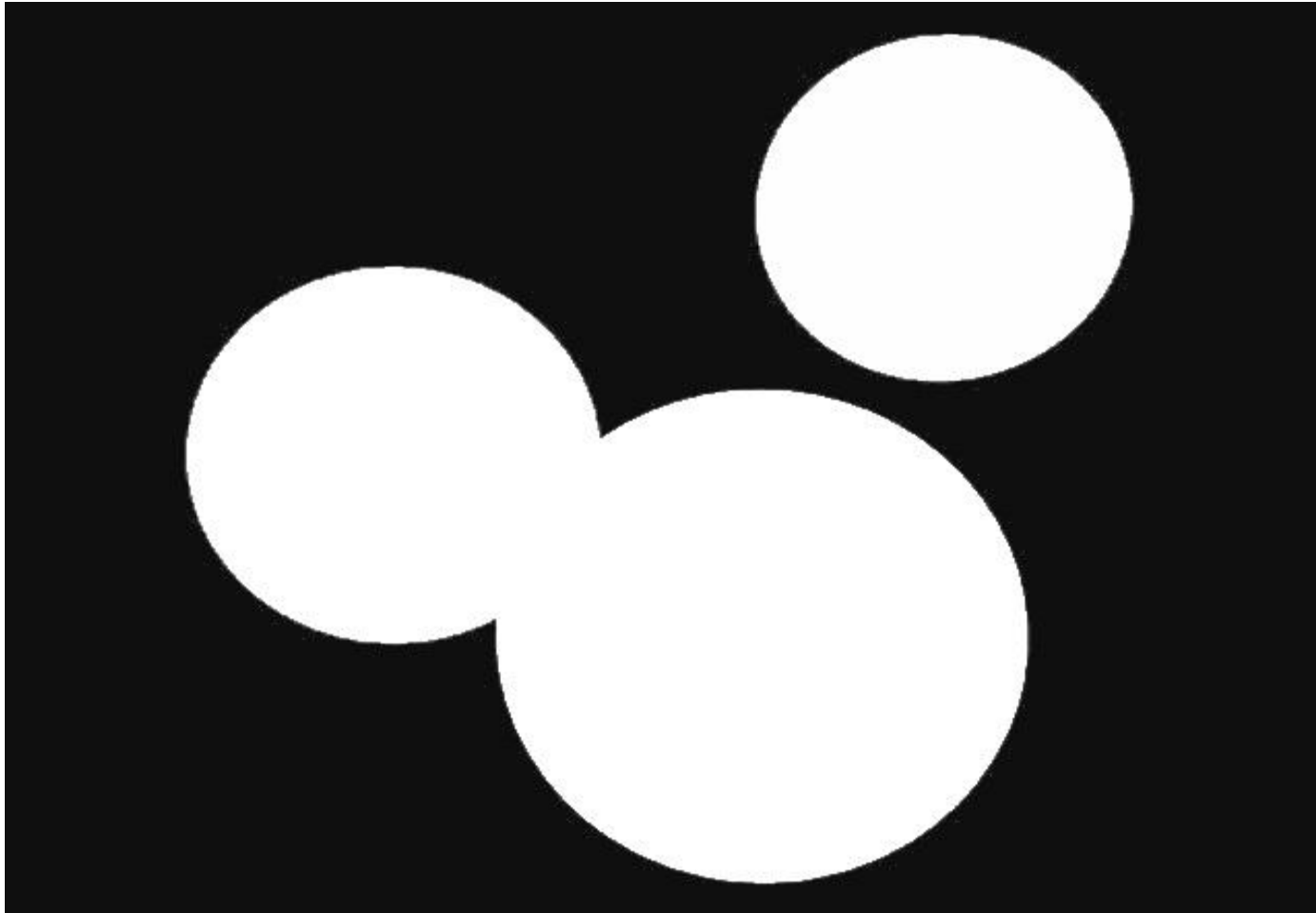


A decorative graphic consisting of a thick gray vertical bar on the left side of the slide and a thin gray horizontal line extending across the top.

Local Illumination

©Alexandre Meyer 2002, Anthony Steed 1999-2003

The Image without Lighting



Introduction

■ *Local illumination*

- *Valid for ray-tracing and for Z-Buffer (projection)*
- *Notation*
 - I_r Intensity radiating from the object (What we're looking for)
 - I_i Normalized intensity of the light (Characteristic of the light)
 - K proportion of the light reflected rather than absorbed by the material (Characteristic of the surface; varies with light wavelength)
- *3 wavelengths: Red, Green & Blue*
- *Illumination: Ambient + Diffuse + Specular*

Ambient Light

- *Approximation to global illumination*
 - *Each object is illuminated to a certain extent by “stray” light*
 - *Constant across a whole object*
- *Often used simply to make sure everything is lit, just in case it isn’t struck by light direct from a light source*

Ambient Light

- *Ambient light usually set for whole scene (I_a)*
- *Each object reflects only a proportion of that (k_a)*
- *So far then* $I_r = k_a I_a$

Lighting Equation #1

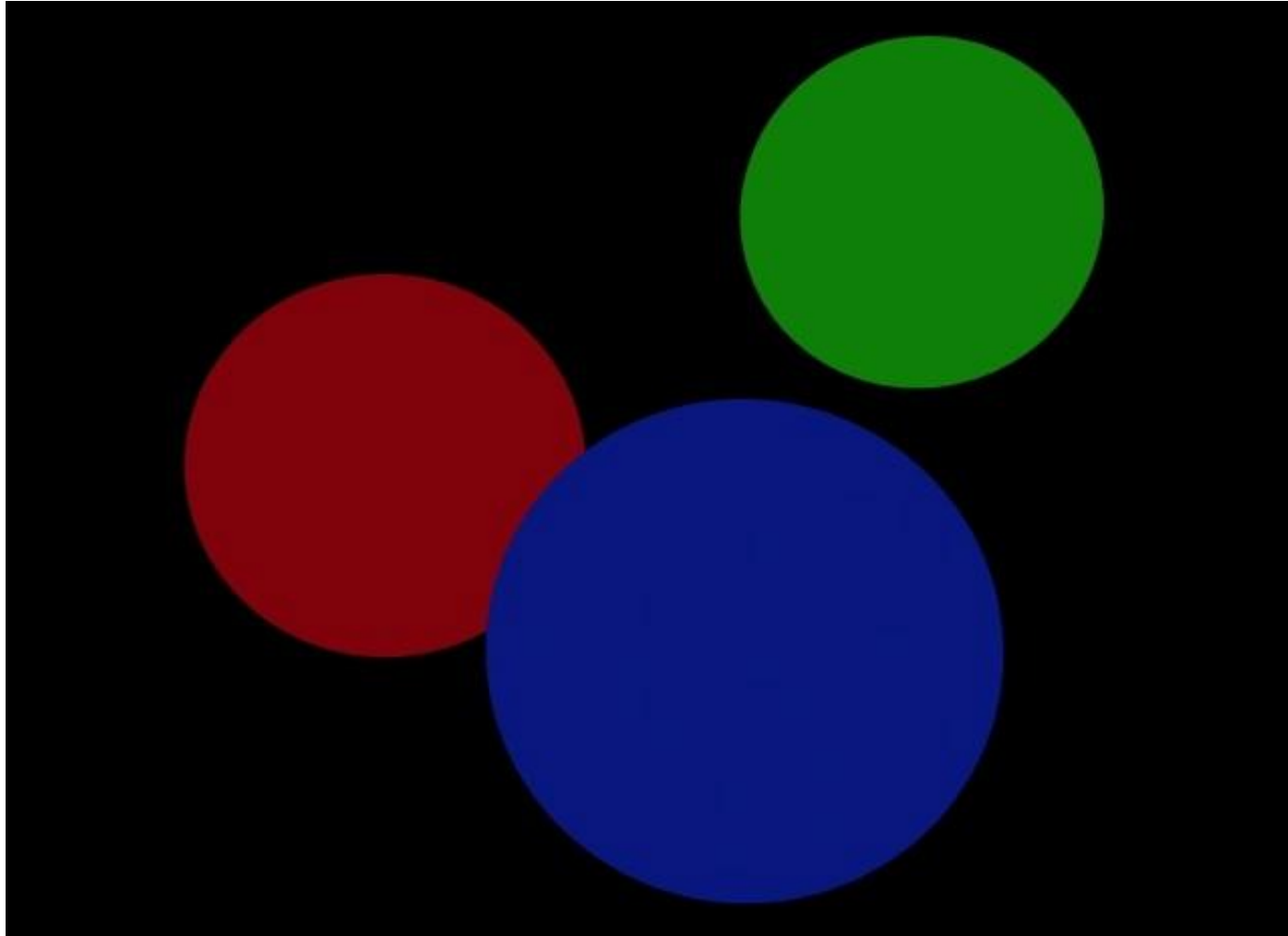
But we use RGB so

$$I_{r, red} = k_{a, red} I_{a, red}$$

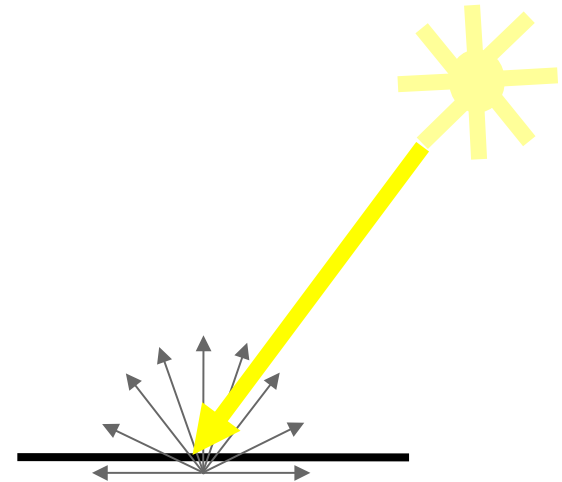
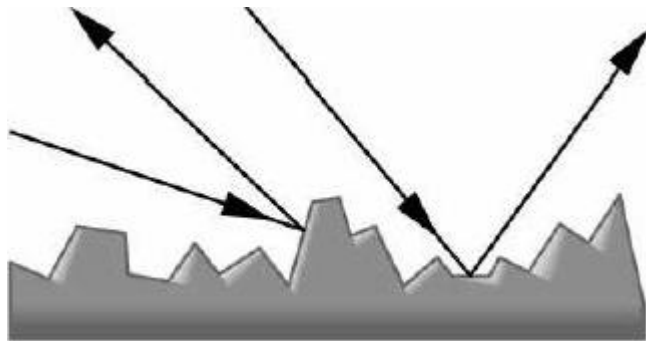
$$I_{r, green} = k_{a, green} I_{a, green}$$

$$I_{r, blue} = k_{a, blue} I_{a, blue}$$

The Image - Ambient

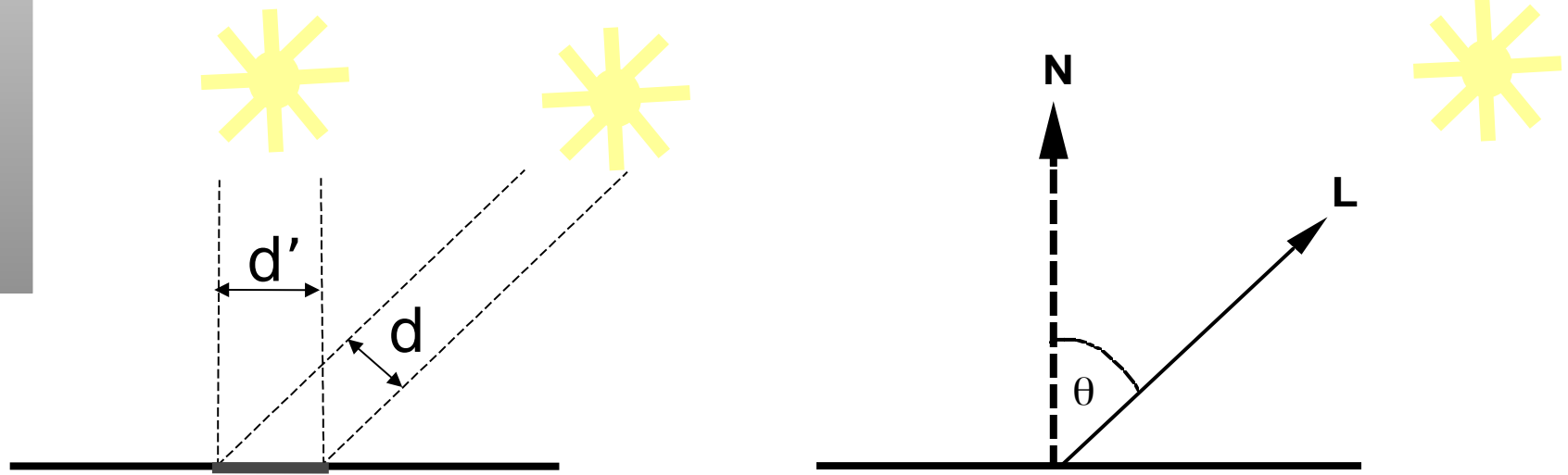


Lambert's Law



- *Diffuse reflector scatters light*
- *Assume equality in all direction*
- *Called Lambertian surface*
- *Angle of incoming light is still critical*

Lambert's Law



- L is the direction to the light
- N is the surface normal

- *Incoming intensity of light is proportional to d*
- *d is proportional to $\cos \Theta = N \cdot L$*

➔ *Reflected intensity is proportional to $\cos \Theta$*

Diffuse Light

- *The normalised intensity of the light incident on the surface due to a ray from a light source*
- *The light reflected due to Lambert's law*
- *The proportion of light reflected rather than absorbed (k_d)*

Lighting Equation #2

$$I_r = k_a I_a + k_d I_i \text{ (n.1)}$$

- *Ambient and diffuse components*
- *Again k_d is wavelength dependent and we work with $k_{d,red}$ $k_{d,green}$ and $k_{d,blue}$*

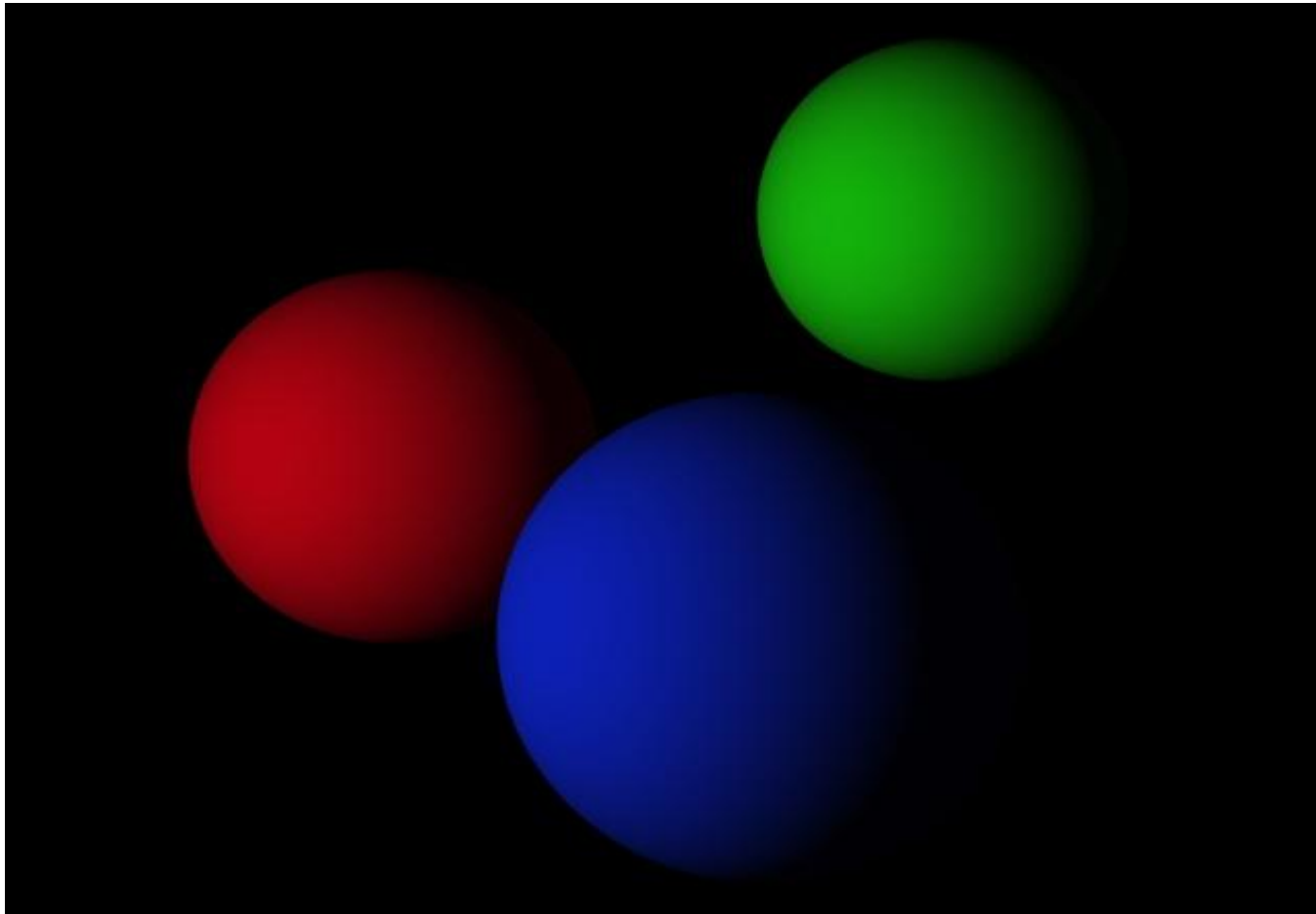
Multiple Lights?

- *Add the diffuse terms*

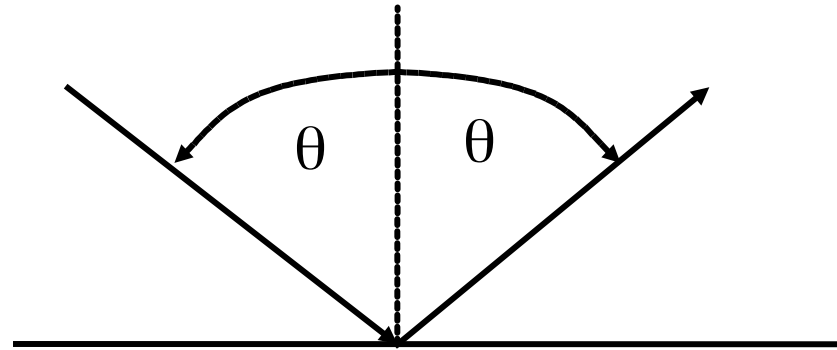
$$I_r = k_a I_a + \sum_{j=1}^m k_d I_{i,j} (n \cdot l_j)$$

- *$I_{i,j}$ is the incoming intensity of light j*
- *l_j is the vector to light j*

The Image - Diffuse

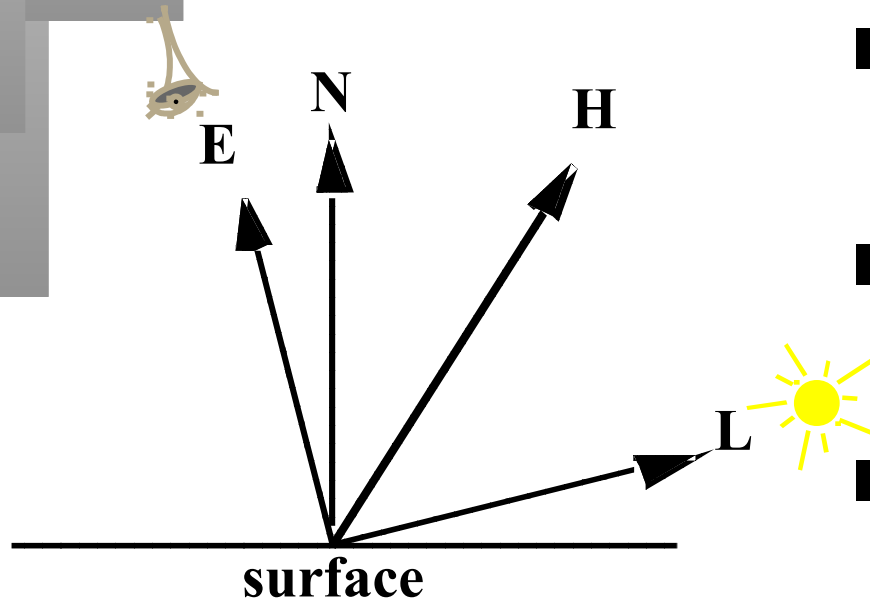


Perfect Specularity



- *Would almost never see the specular highlight*

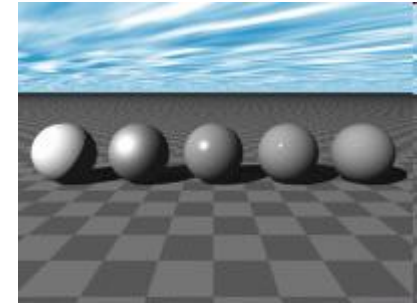
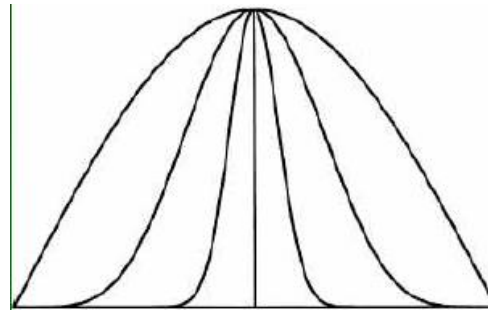
Imperfect Specularity (Phong)



- *E is the direction to the eye*
- *N is the normal*
- *L is the direction to the light*
- *H bisects E and L*

Specular Component

$$k_s I_i (h \cdot n)^m$$



- *m is the power of the light (shininess)*
 - *High m implies smaller specular highlight*
 - *Low m makes the highlight more blurred*

Lighting Equation #3

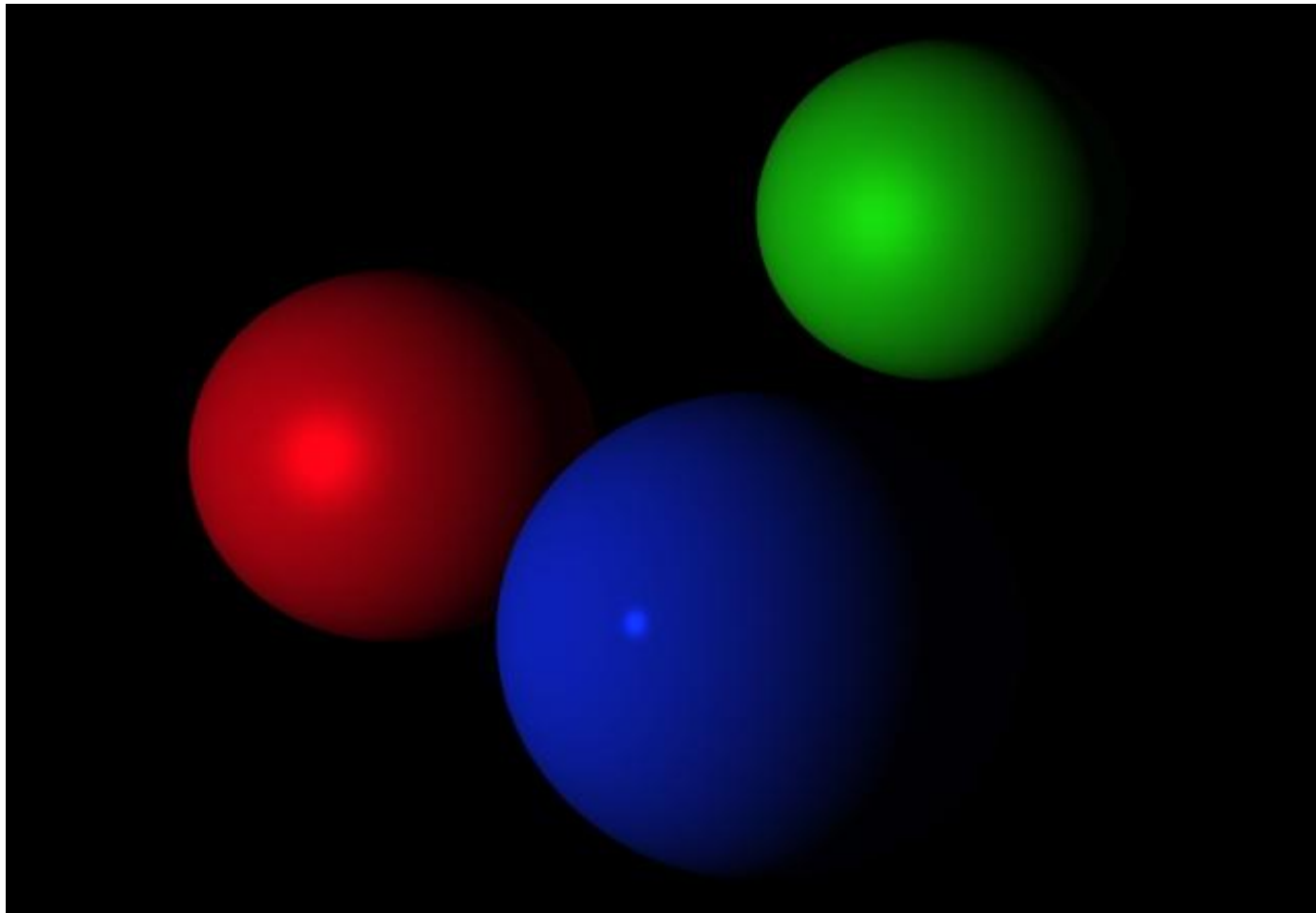
$$I_r = k_a I_a + I_i (k_d (n.l) + k_s (h.n)^m)$$

- *Ambient, diffuse & specular components*
- *Again if there are multiple lights there is a sum of the specular and diffuse components for each light*

(This is the time to worry about clamping values to 0,1 required for monitor display)

The Image - Specular

Small and big specular highlight



Conclusions

- *We can now colour the pixels by combining*
 - *Ambient light*
 - *Diffuse reflections*
 - *Specular reflections**Summed over several light sources*
- *We need*
 - *Shadows*
 - *Better model for light reflection of the object:
BRDF*
 - *Global illumination*